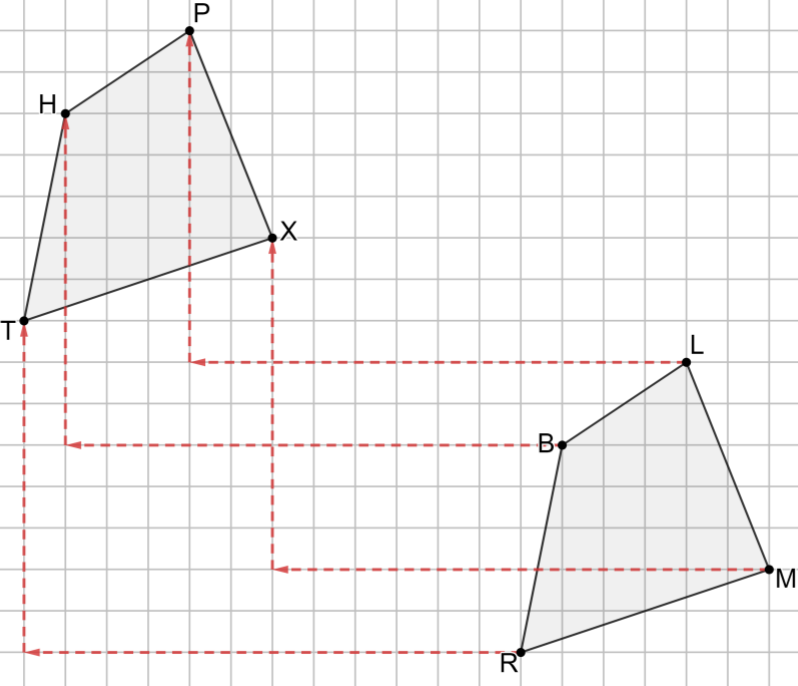


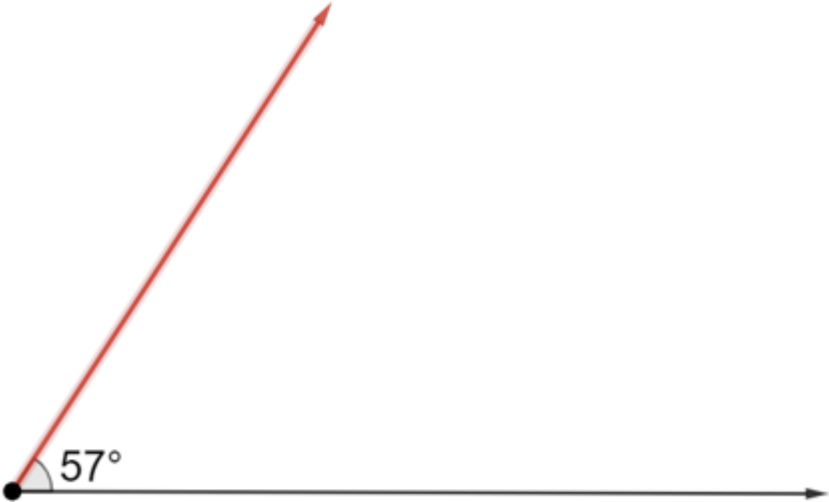
Geometry Unit 1 Assessment Guide

General Information	In unit 1, students are introduced to concepts that build a foundation for the rest of the course. Some of these concepts do not have a directly related benchmark, but the authors of the curriculum included them in this unit as foundations to the course that are not necessarily at the depth of a benchmark or directly associated with a benchmark. These concepts will serve both the teacher and the student as benchmarks are further explored at their depth in subsequent units. In this unit, these concepts were discovered using transformations (MA.912.GR.2.2) but are not assessed within the boundaries of transformations. Examples of this are questions 6 and 7 where the student is using a description of a geometric term. Other examples are questions 11 and 12. “Instruction includes the use of hands-on manipulatives and geometric software for students to explore relationships, postulates and theorems.” <i>From the Geometry B.E.S.T. Instructional Guide for Mathematics (B1G-M) page 14.</i> Unit 1 uses patty paper to explore relationships and to build an understanding of definitions. Question 1 connects a student understanding of rigid motions and idea of a plane being the vehicle of the transformation. Teachers may want to use this assessment to gain a clearer picture of a student’s ability to solve equations. If students are struggling solving equations, teachers are encouraged to support students using grade 7 unit 10 and/or grade 8 unit 8. From the B1G-M, “Instruction includes the connection to compass and straight edge constructions and how the validity of the construction is justified by a proof.” Questions 4 and 11 build off the conjectures that students made while constructing a segment bisector and an angle bisector. These are also foundational to MA.912.GR.1.1, where students apply postulates, relationships and theorems of lines and angles to solving problems. This benchmark will be explicitly addressed in Units 2 and 3. While a student could answer question 11 using a conjecture about the points on a perpendicular bisector of a line segment being equidistant from the segment’s endpoints, the student can also answer using their understanding of a bisector. The relationships of both an angle bisector and a segment bisector are not found outside the benchmark MA.912.GR.5.2. Unit 1 uses the constructions of MA.912.GR.5.2 to build a student’s algebraic foundation which will become important in later units. Teachers may also choose to shorten the assessment to gain certain insight on student understanding of basic prerequisite knowledge that will inform their teaching for other units.
Calculator Information	The questions on the unit assessment are calculator neutral. Teachers can determine whether to allow their students to use the calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.2.2
	Patty paper is similar to a ○ line. ○ point. ○ plane. Drawing a figure on patty paper and then turning the patty paper represents a ○ reflection ○ rotation ○ translation of the figure. Drawing a figure on patty paper and then moving the patty paper to the right and up represents a ○ reflection. ○ rotation. ○ translation. Drawing a figure on patty paper and then flipping the patty paper over represents a ○ reflection. ○ rotation. ○ translation. Reflections, rotations, and translations are known as ○ mappings. ○ rigid transformations. ○ non-rigid transformations.	Recommended Scoring (10 Total Points)	Consider giving students 2 points for each correct answer.
Question	2a	Benchmark(s)	MA.912.GR.2.8
	$\frac{6}{10}$	Recommended Scoring (4 Total Points)	Consider accepting equivalent scale factors.

Question	2b	Benchmark(s)	MA.912.GR.2.8
		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	3	Benchmark(s)	MA.912.GR.2.3
		Recommended Scoring (4 Total Points)	Consider giving 1 point for each correctly mapped vertex. Vectors that go directly from one vertex to another are acceptable. This depends on what was taught within each classroom.

Question	4a	Benchmark(s)	MA.912.GR.5.2
$x =$ 8		Recommended Scoring (6 Total Points)	Consider giving students 2 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x .
Question	4b		
$MT =$ 53		Recommended Scoring (3 Total Points)	Consider following student work for 4a to determine if a minor error is carried through, thereby resulting in a different answer. A student who makes a minor error in question 4a but correctly applies that knowledge in question 4b is showing knowledge and should receive full credit.

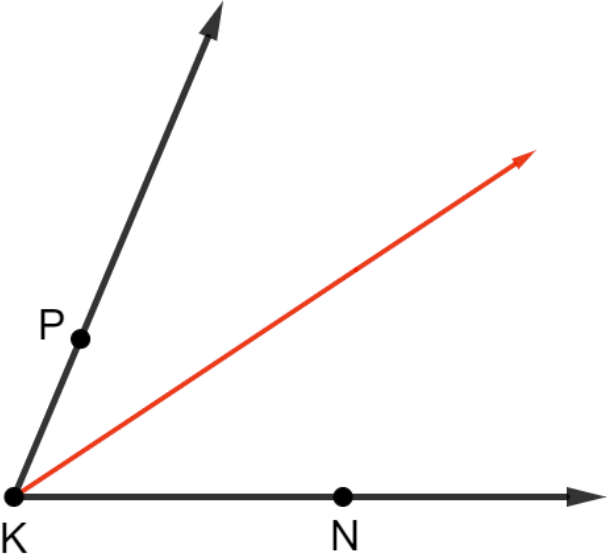
Question	5	Benchmark(s)	MA.912.GR.5.1
		Recommended Scoring (4 Total Points)	Ensure the student has the correct markings to show that the correct steps were used to copy the angle. The angle measure should be close to 57°.
Question	6	Benchmark(s)	<i>Foundational to the course</i>
☐ A defined term ☐ An undefined term	☐ parallel lines ☐ perpendicular lines ☐ skew lines ☐ parallel planes ☐ perpendicular planes	Recommended Scoring (4 Total Points)	Consider giving students 2 points for each answer.

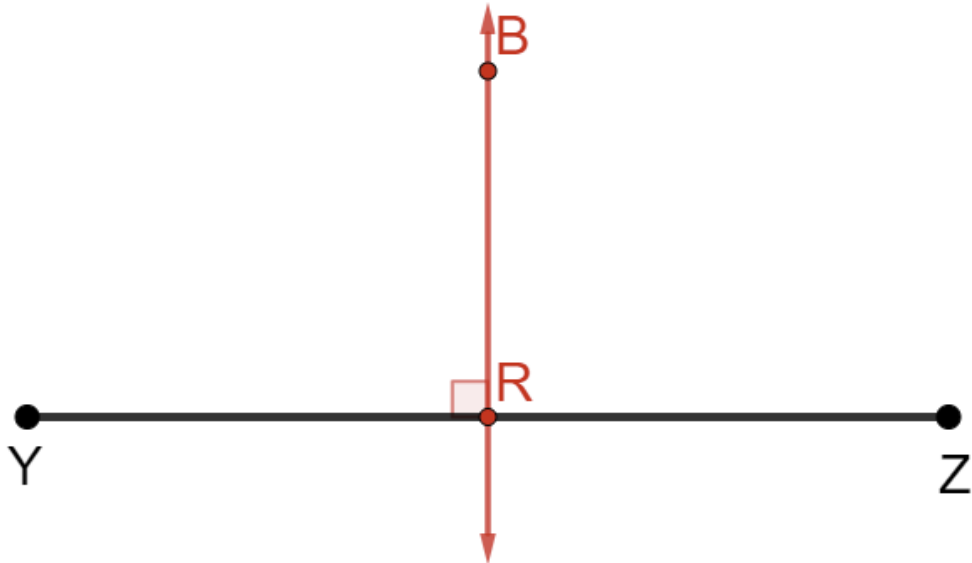
Question	7	Benchmark(s)	Foundational to the course		
<table><tr><td> - A defined term - An undefined term </td><td> - line - plane - skew planes - straight angle - perpendicular lines </td></tr></table>		- A defined term - An undefined term	- line - plane - skew planes - straight angle - perpendicular lines	Recommended Scoring (4 Total Points)	Consider giving students 2 points for each answer.
- A defined term - An undefined term	- line - plane - skew planes - straight angle - perpendicular lines				
Question	8a	Benchmark(s)	MA.912.GR.2.3		
Reflection across line EF .		Recommended Scoring (4 Total Points)	Consider giving students who state that the transformation is a translation of left 6 units 3 points. Within this unit, student understanding of congruence is in its early stages so students may miss the need to be precise.		

Question		8b	Benchmark(s)	MA.912.GR.2.3
Translate left 5 down 5			Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question		8c	Benchmark(s)	MA.912.GR.2.3
Rotation 90° clockwise about point <i>F</i>			Recommended Scoring (4 Total Points)	Consider giving students 2 points if the center of rotation is not stated. Consider giving students 2 points if the student states the rotation is counterclockwise.
Question		9a	Benchmark(s)	MA.912.GR.2.3
Type of Transformation ☐ Dilation ☐ Rotation ☒ Reflection ☐ Translation	Center, Line, or Movement ☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is <u>HN</u> . ☐ Down ☐ Left ☐ Right ☐ Up	Degrees or Scale Factor	Recommended Scoring (4 Total Points)	Consider giving students 2 points for each answer.

Question		9b-d	Benchmark(s)	MA.912.GR.2.3
b. $\overline{XW} \cong \overline{MR}$ c. $\angle GWX \cong \angle MRK$ d. $\overline{WG} \cong \overline{KR}$			Recommended Scoring (12 Total Points)	Consider giving students 4 points for each answer. Consider giving students 3 points if the line segment or angle mark is not provided.
Question		10	Benchmark(s)	MA.912.GR.2.3
Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor	Recommended Scoring (4 Total Points)	Consider giving students 2 points for the type of transformation, 1 point for the center of dilation, and 1 point for the scale factor. Students may give the scale factor as $\frac{4}{6}$.
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is <i>C</i> . ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	$\frac{2}{3}$		
Question		11a	Benchmark(s)	Foundational to MA.912.GR.1.1
$y =$ 13			Recommended Scoring (6 Total Points)	Consider giving students 2 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of y .

Question	11b	Benchmark(s)	Foundational to MA.912.GR.1.1
KY = 216		Recommended Scoring (3 Total Points)	Consider following student work for 11a to determine if a minor error is carried through, thereby resulting in a different answer. A student who makes a minor error in question 11a but correctly applies that knowledge in question 11b is showing
Question	12a		
x = 3.5		Recommended Scoring (6 Total Points)	Consider giving students 2 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x.

Question	12b	Benchmark(s)	<i>Foundational to MA.912.GR.1.1</i>
$m\angle PKN =$ 67°		Recommended Scoring (3 Total Points)	Consider following student work for 12a to determine if a minor error is carried through thereby resulting in a different answer. A student who makes a minor error in question 12a but correctly applies that knowledge in question 12b is showing knowledge and should receive full credit.
Question	13	Benchmark(s)	MA.912.GR.5.2
		Recommended Scoring (4 Total Points)	Ensure the student has the correct markings to show that the correct steps were used to bisect the angle.

Question	14	Benchmark(s)	MA.912.GR.5.2
		Recommended Scoring (4 Total Points)	Ensure the student has the correct markings to show that the correct steps were used to construct the perpendicular bisector.